Highlights

Domain-Validity-Gated Metamorphic Relation Identification and Executable Test Assets for Scientific Machine Learning

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- A domain-validity rubric screens candidate MRs for SciML surrogate testing.
- MR cards convert retained candidates into auditable executable test assets.
- Relation-level verdict ledgers separate pass, fail, skip, OOD, and inconclusive cases.
- MeshGraphNets cylinder flow is used as a case study, not as the paper's main novelty.

Domain-Validity-Gated Metamorphic Relation Identification and Executable Test Assets for Scientific Machine Learning

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Abstract

Context: Scientific machine-learning (SciML) surrogates are increasingly used to approximate expensive physical simulations. Mesh-based neural simulators are attractive for fluid-flow problems because they operate on irregular meshes and predict spatiotemporal physical fields. Testing such systems is difficult because exact expected outputs are often unavailable for arbitrary inputs without running a trusted high-fidelity solver. Rollout accuracy and physics residuals provide useful evidence, but they do not by themselves specify which physical relations should hold under controlled transformations of inputs, meshes, boundary conditions, or parameters.

Objective: This paper investigates how physically meaningful metamorphic relations (MRs) can be screened and operationalized as executable oracle-free test assets for scientific machine-learning systems. The goal is not to improve a predictive model, but to make the step from candidate relation to executable test asset more systematic, auditable, and explicit about the physical, numerical, and software conditions under which each relation is expected to hold. MeshGraphNets-family cylinder-flow surrogates are used as the concrete case study.

Method: We propose a domain-validity rubric for screening candidate MRs, an MR-card and executable-asset format that records source cases, follow-up transformations, output mappings, metrics, tolerances, exclusion rules, and relation-level verdicts, and a case-study protocol for applying these assets to mesh-based neural cylinder-flow surrogates. Candidate sources may include physical equations, boundary conditions, representation contracts, expert reasoning, LLM-assisted candidate lists, and NOETHER-style pattern organization, but validity is decided by the rubric and evidence records rather

than by candidate generation alone.

Results: Full cross-SUT results remain blocked pending cross-SUT artifacts. Only strictly scoped, within-SUT pilot evidence on a single MeshGraphNets checkpoint is reported: node-permutation equivariance holds to machine precision; an approximate mirror-y OOD-stress probe, with the exact relation out of relation domain for this asymmetric mesh, failed on 10 of 10 recorded eval frames (median relative L2 0.737, median V/floor 3.96); and an absolute mass-conservation relation stays deferred while a reference-relative divergence diagnostic passes. Two further within-SUT runs sharpen the interpretation: a same-trajectory one-step rollout-accuracy diagnostic gives median relative L2 0.0216, so the mirror-y violation is about 34 times the surrogate's in-distribution accuracy; and on a synthetic mesh that is provably symmetric about the centerline, where the rubric admits the exact mirror-y relation rather than downgrading it, the surrogate still violates exact mirror-y equivariance (relative L2 1.10), removing the out-of-relation-domain objection to the mirror-y finding. Used as detectors against a 10-mutant injected-fault catalogue, the MRs catch 5 of 10 faults and, in a suggestive, single-SUT test, localize them by MR class. To address a single-checkpoint objection, the five within-SUT measurements are replicated across a roster of K=6 MeshGraphNets cylinder-flow checkpoints (four base-config seed replicas plus one wider and one deeper configuration variant, all from the same read-only trainer on the same DeepMind dataset): node-permutation equivariance is exact on every checkpoint; mirror-y OOD-stress median relative L2 has bootstrap 95% CI $[0.74, 0.80]$ over the seed family with the configuration variants within the same band; the synthetic symmetric-mesh exact-mirror-y test fails on every checkpoint; the reference-relative conservation ratio sits in CI $[1.007, 1.011]$; and the one-step rollout median relative L2 sits in CI $[0.0217, 0.0224]$, so the mirror-y violation is about 35$\times$ the in-distribution accuracy on every checkpoint. The P1 discrete-divergence operator that underwrites the continuity-MR admissibility predicate is empirically $O(h)$ on the symmetric-mesh family (log–log slope 0.988, $R^2 = 1.000$), confirming the numerical-decidability gate has a measurable basis. This is bounded within-SUT and within-architecture-family evidence, not a reliability, accuracy, baseline, cross-architecture-family, geometry-independent, or general fault-detection-rate claim.

Conclusion: The evidence supports a validity-aware bridge from candidate MR ideas to auditable SciML test assets. The scoped pilots show that a rubric-gated workflow can produce relation-level outcomes while pre-

serving the evidence boundary for each claim. We frame this as a domain-admissibility-gated, relation-indexed approach to SciML OOD validation: a relation is admitted only when its tolerance dominates the relevant numerical error floor, and its verdict separates a model-level violation from an out-of-domain application.

Keywords: Metamorphic testing, metamorphic relation identification, oracle problem, scientific machine learning, MeshGraphNets, cylinder flow, software verification and validation

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1. Introduction

Scientific computing increasingly uses learned surrogates to reduce the cost of repeated numerical simulation. In fluid dynamics, mesh-based neural simulators such as MeshGraphNets learn to predict physical fields on irregular meshes and can provide fast rollout predictions for benchmark problems such as flow around a cylinder (Pfaff et al., 2021). These models are useful because they can approximate high-cost solvers, but they also create a familiar software testing problem in a new setting: for many candidate inputs, there is no cheap and exact expected output against which the program can be checked. This is a form of the test oracle problem (Barr et al., 2015).

The common response is to evaluate a surrogate on held-out trajectories and report rollout error. This is necessary, but it is not the whole validation problem. A model may have acceptable average error on a finite test set while still failing to preserve relations that should hold under a physically meaningful transformation. Conversely, a high error value does not always explain which physical, numerical, or representational condition has been violated. For a user who wants to deploy a SciML surrogate outside a narrow validation set, the practical question is not only "How accurate is the model on these samples?" but also "Under which transformations or regimes does this system stop respecting the relations it should respect?"

Metamorphic testing (MT) addresses the oracle problem by checking relations among multiple executions instead of requiring an exact output for each individual test case (Chen et al., 1998; Segura et al., 2016). A metamorphic relation states that if a source input is transformed in a specified way, the corresponding outputs should satisfy a necessary relation. This is a natural idea for scientific computing: conservation laws, symmetry relations, nondimensional similarity, boundary constraints, continuity conditions, and numerical stability expectations all suggest possible relations among executions (Kanewala and Bieman, 2019; Lin et al., 2020; Olsen et al., 2019).

The difficult part is deciding which candidate relations are valid and executable for a particular SciML program. A transformation that looks plausible at a high level may violate the governing assumptions, boundary conditions, discretization choices, or measurement tolerance of the actual system under test (SUT). For cylinder flow, mirror symmetry depends on geometry and boundary labels; translation invariance depends on how coordinates and domain boundaries are represented; Reynolds–Strouhal similarity depends on nondimensionalization and flow regime; a divergence check depends on a discrete divergence operator, mesh weights, and boundary treatment. Treating such transformations as automatically valid MRs would make the test suite look rigorous while hiding the assumptions that determine whether a violation is meaningful.

Prior studies show that relation-based and residual-based evidence is useful when direct oracles are limited in scientific software, simulation testing, design-assumption MRs, and SciML reliability (Hiremath et al., 2021; Mandrioli et al., 2025; Gopakumar et al., 2025). Recent candidate leads also suggest that physics-based MRs are being explored for learned physical-field or fluid-velocity predictors, but the currently verified record is not strong enough to support a first-or-only novelty claim. The remaining problem addressed here is therefore narrower: how candidate MRs are screened for domain validity, turned into executable test assets, and reported through relation-level verdicts that distinguish SUT inconsistency from out-of-relation-domain cases and numerical tolerance effects.

This paper treats MR identification for SciML as a validity-gated testing problem. Physical knowledge, expert reasoning, LLM-assisted candidate lists, and NOETHER-style pattern organization (Li et al., 2026) can all suggest candidate relations, but a candidate relation is not yet an executable MR. It must first state the physical or software basis of the relation, the transformation preconditions, boundary-condition compatibility, output mapping, metric, tolerance rationale, exclusion rule, and verdict interpretation. Only retained relations are converted into executable MR assets.

Two ideas organize this treatment. First, a candidate relation is *admissible* only when, in addition to a physical or software basis and satisfied transformation preconditions, it is numerically decidable: its verdict tolerance must dominate the intrinsic error floor of the operator that measures it — machine precision for an exact representation relation, the interpolation or mapping floor for an approximate geometric relation, or the discretization floor of a discrete operator for a continuity relation. Second, a relation-level verdict is read in two dimensions — how far the measured quantity violates the relation, and how far the transformed case lies outside the relation's validity domain — so that a model-level inconsistency is not confused with a relation applied outside its domain. We refer to this as a domain-admissibility-gated, relation-indexed approach to SciML OOD validation: an organizing framework, not a new model and not a claim of superiority over uncertainty quantification.

The case study is scoped to MeshGraphNets-family cylinder-flow surrogates. This is an intentionally narrow empirical setting rather than the paper's main conceptual contribution. It allows us to examine transformations over meshes, geometry, velocity fields, nondimensional quantities, and rollout behavior while keeping the SUT family concrete enough for reproducible testing. Full cross-SUT evaluation remains blocked. The current case study reports only one trained SUT and checkpoint, so we do not claim external validity across all neural fluid surrogates.

1.1. Research questions

The main research question is:

RQ0. How can candidate metamorphic relations for scientific machine-learning systems be screened for domain validity and converted into executable oracle-free test assets without relying on exact per-sample expected outputs?

We decompose this into four questions:

RQ1 – Validity. How can a domain-validity rubric distinguish physically meaningful candidate MRs from transformations that are executable but invalid, underspecified, or outside the relation's domain?

RQ2 – Operationalization. How can retained candidates be represented as MR cards and executable assets with source cases, follow-up transformations, metrics, thresholds, exclusions, and verdict rules?

RQ3 – Verdict. How can relation-level verdicts distinguish pass, fail, skip, out-of-relation-domain, numerical tolerance issue, and inconclusive outcomes?

RQ4 – Case-study evidence. In a MeshGraphNets-family cylinder-flow case study, what evidence does the rubric-gated asset workflow add relative to expert MR design, LLM-assisted candidate generation, generic MR-generation scope contrasts, and rollout-accuracy diagnostics? In the current evidence only the rollout-accuracy comparator is answered; the expert-MR, LLM-candidate, and generic-MR comparators remain planned and are blocked pending their artifacts.

1.2. Contributions

This paper makes three scoped contributions, each repositioned narrowly with respect to the closest prior we identify in Section 2.4, and evaluated through a single MeshGraphNets-family cylinder-flow case study.

First, we propose a domain-validity rubric and MR-card executable-asset workflow that screens candidate MRs and converts retained ones into auditable, executable test assets recording physical basis, transformation preconditions, boundary-condition compatibility, output mapping, metric, tolerance, and exclusion rules. As a SciML-specific instantiation of the calibrated-tolerance principle introduced by Eniser et al. (Eniser et al., 2022) as *relaxations* for stochastic RL policy testing, we ground the tolerance floor in the *intrinsic error floor of the discrete measurement operator*, whose truncation order is theoretically characterizable (for a P1 discrete-divergence operator on a triangular mesh, $O(h)$ in the mesh spacing); in this study the floor magnitude is estimated from the reference field on the deployed mesh rather than computed from a closed-form bound, which we leave to future work. NOETHER-style pattern organization may serve as one candidate source but does not decide MR validity. To our knowledge, grounding an MR tolerance in the measurement operator's own error floor in this way is new in the SciML setting.

Second, we define a relation-level verdict and ledger scheme that reads verdicts in two dimensions (relation-violation against domain-violation magnitude). This instantiates, for physics-governed SciML, the constraint-architecture pattern introduced by Duque-Torres et al. (Duque-Torres et al., 2023a), formalised by *Towards a Complete Metamorphic Testing Pipeline* (Duque-Torres et al., 2023c), and automated data-drivenly by MetaTrimmer (Duque-Torres et al., 2023b); what is new here is a *typed classification* of domain-inadmissibility sub-dimensions drawn from PDE-domain preconditions, geometry, boundary conditions, and operator admissibility.

The relation-violation axis is quantified; the domain-violation axis is at present only qualitatively operationalized, and a calibrated continuous score is left to future work.

Third, as an element that none of the closest prior works addresses, we use the paper's own MRs as fault detectors against an independently re-implemented 10-mutant seeded-fault catalogue, and report a by-class localization: the continuity MR localizes to boundary-condition and gross normalization-scale faults; the symmetry MR localizes to physical-channel and mesh-adjacency faults; node-permutation equivariance, by-design exact under these faults, localizes to none. This is suggestive evidence for a typed mapping from MR class to fault class, not a validated localization model.

We evaluate these contributions on one trained MeshGraphNets cylinder-flow checkpoint, extending the active-transformation testing direction of Reichert et al. (Reichert et al., 2024) — who applied physics-derived MRs to a trained LSTM hydrologic surrogate and produced a basin-stratified applicability map — from hydrology to mesh-based neural fluid surrogates. The evidence comprises three scoped relation pilots, a same-SUT rollout-accuracy comparator, and an exact mirror-y test on a provably symmetric admissible synthetic mesh; expert MR design, generic MR-generation scope contrasts, LLM-assisted candidate generation, and cross-SUT comparisons remain protocol commitments until matched artifacts exist.

2. Background and Related Work

2.1. Mesh-based neural simulation

Mesh-based neural simulators learn dynamics on graph or mesh representations of physical systems. For fluid-flow problems, a mesh representation is attractive because the geometry, boundary regions, and local connectivity can be represented without forcing the state onto a regular grid. MeshGraphNets is a representative architecture family in this area: it encodes mesh nodes and edges, propagates information by message passing, and predicts future physical fields through autoregressive rollout (Pfaff et al., 2021).

The cylinder-flow benchmark is useful for testing because it combines several features that matter for SciML validation. It involves geometry, boundary conditions, velocity fields, pressure-like quantities or derived quantities, temporal rollout, and flow-regime assumptions. It also exposes the distinction between data-driven accuracy and physical relation preservation. A model may fit a trajectory distribution while failing under a controlled transformation of geometry, mesh representation, or flow parameters.

In this paper, MeshGraphNets-family systems are treated as software systems under test. We do not evaluate them as new modeling contributions. Instead, we ask how their outputs behave under controlled input transformations and whether necessary relations remain within explicit tolerances.

2.2. Metamorphic testing and the oracle problem

Metamorphic testing was developed to address programs for which it is difficult or impossible to know the correct output for a single test input (Chen et al., 1998; Segura et al., 2016). Instead of checking one output against one expected value, MT checks a relation among the outputs of a source input and one or more follow-up inputs. This framing has been applied to scientific software, simulation models, and machine-learning systems, all of which can suffer from oracle problems (Xie et al., 2011; Kanewala and Bieman, 2019; Olsen et al., 2019).

For SciML surrogates, the oracle problem is especially visible in out-of-distribution (OOD) settings. A trusted solver may be too expensive to run for every transformed case, and even when a reference output is available, a single pointwise error does not necessarily indicate whether a physical relation has been preserved. MR-based testing offers a complementary perspective: it asks whether the SUT maintains a necessary relation under a controlled transformation.

However, SciML MRs cannot be treated as generic input perturbations. In image-based ML testing, transformations such as lighting or weather changes may preserve the semantic label. In scientific computing, the transformation must respect governing equations, physical regimes, boundary conditions, mesh representation, and numerical tolerance. The correctness of the MR itself becomes a central validity question.

2.3. MR identification for scientific and simulation software

Prior work on scientific software testing has shown that MRs can be elicited from monotonicity, conservation-like behavior, scaling relations, simulation assumptions, and domain-specific expectations (Kanewala and Bieman, 2019; Lin et al., 2020). Work on simulation validation similarly treats multi-run relations as pseudo-oracles when direct validation data are limited or unavailable (Olsen et al., 2019; Raunak et al., 2021). These studies establish that MR thinking is practical for scientific and simulation software, not merely for toy functions.

Other work has explored automated or semi-automated MR identification. Research on ocean system models shows that data-driven search can discover candidate transformations in complex physical software (Hiremath et al., 2021). CPS testing work shows that design assumptions can be encoded as MRs and then used to generate new test traces (Mandrioli et al., 2025). These ideas are important for the present paper because they show that MR sources can include system assumptions, model structure, and physical design knowledge.

The gap is that candidate identification is not enough for SciML. A candidate relation must also state when it is supposed to hold. Without a domain-of-validity record, a failed test may mean several different things: the relation was invalid under the chosen boundary conditions, the transformation left the physical regime, the tolerance was not numerically justified, or the SUT violated a relation it should have preserved. This paper makes that distinction explicit.

2.4. Physics-based MT for learned scientific simulators

Three contemporary works are the closest prior to ours and frame what is and is not new here. We engage each one explicitly.

Reichert et al. (2024) (Reichert et al., 2024) applied physics-derived metamorphic relations to a trained LSTM hydrologic surrogate, perturbing climate-forcing inputs in directions whose physical response is known a priori, stratifying pass/fail by basin elevation to produce an implicit applicability map, and excluding basins where forcing uncertainty dominated the response signal — an informal admissibility filter. We formalise these practices as an explicit admissibility predicate, a two-dimensional verdict type, and a relation-indexed applicability map, and extend them from hydrology to mesh-based neural fluid surrogates whose outputs live on irregular meshes and whose MR validity depends on geometry and discrete operators rather than basin physiography. We do not claim to be the first to use physics-derived MRs on a trained neural surrogate.

Eniser et al. (2022) (Eniser et al., 2022) introduced *relaxations* — numerical tolerances embedded inside MR oracles — for action-policy testing of stochastic reinforcement-learning systems, deriving them empirically from policy rollouts. We extend this calibrated-tolerance principle to deterministic numerical surrogates by grounding the tolerance floor in the intrinsic error floor of the discrete measurement operator, whose truncation order is theoretically characterizable (for a P1 discrete-divergence operator on a triangular mesh, $O(h)$ in the mesh spacing). The distinction from rollout-derived relaxations is that the floor is a property of the measurement operator rather than of model stochasticity; in the present study we estimate its magnitude empirically from the reference field on the deployed mesh, and a closed-form a-priori bound is left to future work. We do not claim that calibrated MR tolerance is itself new.

The 2023 violation-attribution cluster — Duque-Torres et al. (SANER 2023) (Duque-Torres et al., 2023a), *Towards a Complete Metamorphic Testing Pipeline* (Duque-Torres et al., 2023c) and MetaTrimmer (Duque-Torres et al., 2023b) — identified the bug-vs-MR-inapplicability separation as a research problem and addressed it architecturally with explicit MR constraints used as a pipeline pre-filter, with MetaTrimmer automating the constraint derivation from random-input violation logs. Our two-dimensional verdict instantiates this architectural pattern for physics-governed SciML; what is new is a *typed classification of domain-inadmissibility sub-dimensions* drawn from PDE-domain preconditions, geometry compatibility, boundary-condition compatibility, and operator admissibility, rather than a binary skip/proceed gate or a data-derived constraint set. We do not claim that the bug-vs-inapplicability separation is itself new.

Across all three closest works, the element our debates could not pre-empt is the seeded-fault MR-as-detector with by-class fault localization reported in Section 5.3. None of Reichert, Eniser, or the 2023 cluster attempts to map MR failures back to identifiable fault classes within the system under test.

2.5. SciML V&V, residuals, UQ, and failure modes

SciML reliability research has developed several important tools: physics residuals, uncertainty quantification, conformal prediction, certified error bounds, stress testing, equivariance error, and failure-mode analysis. PINNs and neural operators provide important examples of learned systems for PDE-governed problems (Raissi et al., 2019; Karniadakis et al., 2021; Li et al., 2021). Work on PINN failure modes shows that physical constraints in a training loss do not guarantee reliable behavior in more difficult regimes

(Krishnapriyan et al., 2021). Calibrated physics-informed uncertainty quantification uses PDE residuals as nonconformity scores and provides statistical coverage guarantees (Gopakumar et al., 2025).

These methods are complementary to MT. A residual, conservation error, or equivariance error can be a useful diagnostic, but it is not automatically an MR. It becomes part of an executable relation only when there is a source case, a follow-up transformation, an expected output relation, a tolerance rule, and a verdict interpretation. This distinction is central to our paper. We use SciML diagnostics as possible relation measurements, while MT supplies the multi-execution oracle-free structure.

2.6. Hybrid ML-solver trust regions

Hybrid ML-solver frameworks provide another relevant line of work. Some systems use residual thresholds or error estimates to decide when a learned component should be trusted and when a numerical solver should take over (Baral et al., 2026; Wang et al., 2025). Such systems operationalize a form of trust region, although often at runtime and through residual-based switching rather than offline relation-based testing.

Our study pursues a complementary direction. Before deployment, physically derived transformations are used to estimate where relation violations occur and what regimes, boundary conditions, or numerical assumptions are associated with them. The result is not a runtime switching policy by itself, but an evidence structure that can support later decisions about when a learned surrogate should be trusted.

The distinction from residual- and uncertainty-based trust estimation is deliberate. Uncertainty quantification, conformal prediction, and residual-threshold trust regions locate unreliable behavior in feature, residual, or error-estimate space, and they do so passively from observed inputs. The present method instead acts in relation space: it applies a physically derived, controlled transformation and reports which necessary relation breaks under it, indexed by that transformation. The intended product is therefore the evidence structure needed for a relation-indexed applicability map — a statement of the form "under this controlled transformation, this relation no longer holds" — rather than a scalar field of high residual, and the two-dimensional verdict separates a model-level violation from an out-of-domain application, a separation that, as the present case illustrates, a scalar accuracy or residual magnitude does not by itself provide. Section 5.3 gives that concrete within-SUT instance: a surrogate that is accurate in-distribution (median one-step relative L2 0.0216) still violates mirror-y equivariance by roughly an order of magnitude more, so the accuracy number does not bound the relation violation. We do not claim a completed applicability map in this paper; the mirror-y result reported below is one bounded within-SUT example of the evidence such a map would aggregate.

2.7. *What is new and what is not new*

This paper does not claim that metamorphic testing, MR identification, scientific-software MT, residual diagnostics, uncertainty quantification, LLM candidate generation, or NOETHER-style candidate organization is new. These are established or emerging sources of testing ideas. The narrower claim is that SciML MR identification should be treated as a domain-validity problem: a candidate relation becomes useful only after its physical basis, transformation preconditions, output mapping, tolerance, exclusion rule, executable artifact, and relation-level verdict are recorded.

What is new here is the evidence-gated conversion from candidate relation to executable SciML MR asset. The contribution is not a stronger neural simulator and not an automatic MR generator. It is a workflow that makes the validity boundary inspectable, so that a candidate can be retained, rejected, downgraded to OOD-stress, or deferred instead of being silently treated as a valid oracle. Structurally, the novelty is two organizing devices rather than a checklist of MR fields: an admissibility gate that ties a relation's tolerance to the numerical error floor of its own measurement, and a two-dimensional relation-level verdict that separates a model violation from an out-of-domain application. The admissibility gate is fully operational (it decides retain, downgrade, or defer on stated grounds in the case study); the verdict's domain-violation axis is, so far, only qualitatively operationalized, so the two-dimensional verdict is at this stage an interpretive structure pending a calibrated domain-violation score rather than a fully measured device. Both are means to make the validity boundary inspectable, not claims of empirical superiority over existing diagnostics. A third, empirically distinct element, examined in Section 5.3, is the seeded-fault by-class localization: used as detectors against an independently re-implemented injected-fault catalogue, the MRs map by class to fault class (continuity to boundary/scale faults, symmetry to physical-channel/adjacency faults), an element none of the three closest prior works addresses.

3. Method

3.1. Overview

The proposed method is a five-stage workflow:

(1) identify candidate relation sources;

(2) organize candidate relations using declared candidate-source categories;

(3) screen candidates with a domain-validity rubric;

(4) convert retained relations into executable MR assets;

(5) execute the assets and record relation-level verdicts.

The method deliberately separates candidate generation from validity judgment. NOETHER-style patterns may help organize or generate candidate relation structures, but they do not certify that a relation is physically valid for a particular SUT, dataset, mesh, boundary condition, or numerical tolerance. Validity is determined by the rubric and by executable evidence.

3.2. Candidate relation sources

For cylinder-flow surrogates, candidate MRs may come from six sources.

Physical equations and constraints. Incompressible continuity, boundary behavior, and conservation-like expectations can suggest relation measurements such as discrete divergence or boundary-condition consistency.

Geometric and symmetry assumptions. Mirroring, rigid transformations, and coordinate-frame changes may suggest equivariance or invariance checks, but only under compatible geometry and boundary labels.

Nondimensional similarity. Reynolds-number and Strouhal-number relations may suggest follow-up transformations over flow parameters or extracted wake behavior, but only within regimes where the empirical or theoretical relation is meaningful.

Mesh and graph representation. Node permutations, face ordering, edge encoding, and mesh refinement may suggest representation-level MRs.

Temporal rollout behavior. Autoregressive rollouts may suggest determinism, prefix consistency, or semigroup-like sanity checks. These are useful implementation or numerical-consistency checks, but should not be overstated as physical laws.

Cross-implementation comparison. Outputs from different implementations may support method-comparison checks only if units, state variables, meshes, rollout horizons, boundary conditions, and checkpoints are comparable. Otherwise they are triangulation evidence rather than retained MRs.

3.3. Domain-validity rubric

We treat screening as deciding a single admissibility predicate. A candidate relation is an *admissible MR* when four conditions hold together: (i) it has a physical or software basis; (ii) its transformation preconditions are satisfied; (iii) its boundary conditions and output mapping remain compatible after the transformation; and (iv) it is numerically decidable, meaning the verdict tolerance dominates the intrinsic error floor of the operator that measures the relation. Conditions (i)–(iii) establish that the relation is meaningful for the transformed case; condition (iv) establishes that a violation can be told apart from numerical noise. None of the four is optional and none is merely documentary: each is a gate that can reject or defer a candidate, and provenance recording is the mechanism that makes each gate auditable, not a substitute for it. The six rubric criteria are the recorded, auditable form of these four conditions.

Each candidate MR is screened using the rubric in Table 1. The rubric is intentionally conservative. It can retain a relation as an executable MR, retain it only as an OOD stress relation, reject it, or defer it pending missing evidence.

The rubric outputs one of four screening decisions:

- retained as executable MR;
- retained as OOD stress relation, not physics-preserving MR;
- rejected as invalid or underspecified;
- deferred pending missing evidence, such as a discrete operator or threshold.

The defer and OOD-stress decisions are the two ways the admissibility predicate fails softly. A relation is deferred when condition (iv) cannot yet be established — the tolerance cannot be shown to dominate the measuring operator's error floor, as for an absolute discrete-divergence relation whose reference field already carries non-negligible divergence — and it is downgraded

Table 1: Domain-validity rubric for candidate MRs.

Criterion	Question asked before retention
Physical basis	What equation, boundary condition, nondimensional law, representation property, numerical assumption, or implementation contract justifies the relation?
Transformation preconditions	What exactly changes from source to follow-up, and what must be preserved?
Boundary-condition compatibility	Does the transformation preserve the intended physical meaning of the domain and boundaries?
Output mapping	Can the expected output relation be measured from available SUT outputs?
Metric and tolerance	What metric is used, and where does the tolerance come from?
Failure diagnosability	What can a violation plausibly mean, and what can it not mean?

to an OOD-stress relation when conditions (ii)–(iii) hold only approximately, so the transformed case sits near, but outside, the validity domain.

3.4. MR card and executable asset format

A retained MR is represented as an MR card. The card records the MR identifier, source category, physical or software basis, source-case schema, follow-up transformation, transformation preconditions, boundary-condition compatibility, output mapping, metric, tolerance and provenance, expected verdict classes, exclusion rules, artifact schema, and expected fault or boundary interpretation.

Table 2 gives the initial card skeleton. The entries are not final empirical findings; they specify what evidence each relation must provide before it can be used in the study.

Executable assets are generated from MR cards. Each asset includes transformation code, runner configuration, metric computation, verdict logic, and artifact recording. The asset format is intended to make MR execution auditable: another researcher should be able to inspect why a relation was retained, how the follow-up case was generated, how the comparison was made, and what a violation means.

Table 2: Initial MR card skeletons for the cylinder-flow study.

MR class	Candidate relation	Validity evidence needed before execution
Rep.	Node permutation equivariance	Same graph semantics, unchanged physical fields, output permutation inverse, deterministic adapter.
Rep.	Face-order invariance	Face encoding must not carry physical orientation unless explicitly transformed.
Geometry	Mirror-y equivariance	Symmetric geometry, boundary labels, vector-component mapping, and mesh transformation.
Constraint	Discrete divergence boundedness	Discrete divergence operator, mesh weights, boundary treatment, and tolerance provenance.
Similarity	Reynolds–Strouhal consistency	Nondimensional variables, valid flow regime, extraction window, and stationarity condition.
Temporal	Rollout prefix consistency	Same initial condition, deterministic execution, same horizon convention, and tolerance rule.
Method comp.	Cross-implementation consistency	Comparable variables, units, meshes, checkpoints, rollout horizon, and boundary conditions.

3.5. Worked examples: from candidate relation to executable MR card

This subsection gives three design-time worked examples. They are not empirical findings and do not report whether any SUT passes or fails. Their purpose is to show how the rubric changes candidate relations into executable assets, or prevents a plausible physical idea from being overused.

Table 3 summarizes the screening decisions. The three examples were selected because they stress different parts of the method: representation semantics, physical symmetry under boundary conditions, and a constraint measurement whose validity depends on a discrete operator and tolerance provenance.

MR-1: node permutation equivariance.. The source case is a mesh graph with node features, edge connectivity, boundary labels, and physical fields. The follow-up case is produced by applying a permutation π to node order and the corresponding edge indices, without changing coordinates, fields, units,

Table 3: Worked screening decisions for three candidate MRs.

Candidate	**Main validity issue**	**Screening decision**	**Reason**
Node permutation equivariance	Graph relabeling must preserve the same physical mesh and fields	Retain as executable representation MR	The transformation changes representation order but not the physical case, provided the adapter and inverse output mapping are deterministic.
Mirror-y equivariance	Geometry, boundary labels, and vector components must transform consistently	Retain only under boundary-compatible conditions	The relation is physically meaningful for symmetric cylinder-flow cases, but invalid if the transformation changes inlet, outlet, wall, or obstacle semantics.
Discrete divergence boundedness	The metric depends on the discrete divergence operator, mesh weights, boundary treatment, and tolerance	Defer or retain as qualified continuity-constraint MR	The physical basis is strong for incompressible flow, but the executable verdict is meaningful only when the numerical operator and threshold are justified.

time step, boundary labels, or rollout horizon. The expected relation is equivariance: if the SUT output for the source case is Y, then the output for the permuted case should be $\pi(Y)$, up to numerical tolerance. The output mapping applies π^{-1} to the follow-up output before comparison.

This candidate passes the physical-basis criterion as a representation-level contract rather than a fluid-mechanics law. It passes transformation preconditions only if graph relabeling leaves the physical case unchanged. It passes output mapping because velocity and pressure-like node fields can be inverse-permuted. A suitable metric is the normalized node-field difference after inverse permutation, computed separately for each predicted variable and rollout step. The tolerance should be derived from deterministic repeat runs or machine-precision and framework-nondeterminism measurements, not chosen after observing violations. A failure within the relation's valid domain points first to graph adapter, batching, parser, or message-passing implementation issues; it should not be interpreted as a Navier–Stokes physical violation.

MR-2: mirror-y equivariance.. The source case is a cylinder-flow case whose geometry, mesh, and boundary labels are compatible with reflection about the centerline. The follow-up case is generated by mapping coordinates (x, y) to $(x, -y)$ and mapping vector fields accordingly: the streamwise velocity component is preserved, the transverse component changes sign, and scalar fields are mirrored without sign change. Boundary labels must be transformed with the mesh. The expected relation is that the mirrored follow-up prediction, mapped back to the source coordinates and vector convention, matches the source prediction within tolerance.

This candidate illustrates why a physically plausible symmetry is not automatically a valid MR. It is retained only when the cylinder, channel, inlet, outlet, wall labels, forcing, nondimensionalization, and sampling window remain semantically compatible after reflection. It is rejected for cases where the transformation swaps non-equivalent boundaries, breaks obstacle labeling, or leaves vector components unmapped. The metric can be a weighted field error after inverse mirror mapping, reported for velocity components and any scalar output separately. The exclusion rule should remove cases with asymmetric boundary conditions, incompatible mesh metadata, or transformations that require unavailable fields. A violation can indicate a symmetry inconsistency in the SUT, but it can also indicate that the MR was applied outside its domain. The relation-level verdict must preserve this distinction.

MR-3: discrete divergence boundedness.. The source case is an incompressible cylinder-flow trajectory or rollout state with mesh geometry sufficient to compute a discrete divergence estimate. The follow-up relation can be formulated in two ways. In the simpler single-run form, predicted velocity fields should remain within a bounded discrete divergence tolerance for eligible interior cells or nodes. In a transformation-based MR form, a source and follow-up case that preserve incompressible-flow assumptions should not increase the divergence metric beyond the stated relation threshold after output mapping. In both forms, the executable relation requires a declared discrete operator.

This candidate has a strong physical basis as a continuity constraint, but it is not a Noether-derived conservation-law claim in this paper. It is retained only after the discrete divergence operator, mesh weights, boundary treatment, and tolerance provenance are specified. The output mapping must state which predicted variables are used and how boundary nodes, obstacle-adjacent cells, and missing pressure fields are handled. The metric may be an L_2 or L_∞ divergence norm over eligible regions, possibly normalized by a velocity and length scale. The exclusion rule should skip cases where the mesh does not support the operator, where boundary treatment dominates the metric, or where the flow regime is outside the incompressibility assumption. A failure is therefore interpreted as a continuity-constraint violation only after operator and tolerance evidence rule out a numerical artifact.

These examples show the intended use of the rubric. It does not merely list domain questions; it changes the status of candidate relations. Node permutation becomes a retained representation MR, mirror-y becomes a conditional physical-symmetry MR, and divergence becomes a qualified continuity-constraint MR whose verdict depends on numerical instrumentation. This is the level at which candidate MR identification becomes a SciML validity problem rather than a prompt-generation or intuition-listing problem.

3.6. Relation-level verdicts

An MR execution can produce several verdicts:

pass the relation holds within the stated tolerance;

fail the relation is violated within its valid domain;

skip the case cannot be run because a declared precondition is missing;

out-of-relation-domain the transformation produced a case outside the relation's validity domain;

numerical-tolerance issue the evidence is dominated by threshold or numerical-resolution uncertainty;

inconclusive the artifact is insufficient to decide.

This scheme prevents a common overclaim: not every relation violation is a program fault. A violation may reveal a model inconsistency, but it may also reveal that the MR was applied outside its domain, that the threshold was too tight, that the mesh operator was unsuitable, or that the case is not comparable.

It is useful to read these verdicts as regions of a two-dimensional space rather than as a flat list. One axis is the relation-violation magnitude — how far the measured quantity exceeds the tolerance, for example the violation-to-tolerance ratio or the violation-to-floor ratio V/floor. The other axis is the domain-violation magnitude — how far the transformed case lies outside the relation's validity domain, signalled by precondition violation, geometry mismatch, boundary-condition mismatch, or operator inadmissibility. Low domain-violation with low relation-violation is pass; low domain-violation with high relation-violation is the only region that may be read as SUT inconsistency; high domain-violation is out-of-relation-domain or, near the boundary, OOD-stress; a relation-violation that sits within the error floor is a numerical-tolerance issue. This decomposition is what keeps a model-level violation from being confused with a relation applied outside its domain, and it makes condition (iv) of the admissibility predicate explicit at verdict time.

In the present study the relation-violation axis is quantitative — mirror-y reports V/floor — but the domain-violation axis is so far only partially operationalized. The mirror-y case is placed near the validity boundary *qualitatively*: the reflection is non-bijective and the worst reflected-node mismatch is about one mesh edge length, so the exact relation is out-of-relation-domain and is downgraded to an approximate OOD-stress probe. We do not claim a fully calibrated, continuous domain-violation score; defining such a score across MR classes is left to future work. The two-dimensional reading is used here as an interpretive structure, not as a calibrated boundary measurement.

3.7. Hierarchical interpretation protocol

We organize retained MRs into a three-level hierarchy inspired by MR classification for scientific computing (Yang et al., 2020): physical-model relations, computational-model relations, and code-model relations. For learned mesh surrogates, the computational-model level includes graph representation and message-passing discretization assumptions.

We use this hierarchy as a predeclared interpretation protocol that maps representation-level MRs to possible graph encoding or adapter problems, physical-model MRs to possible continuity, symmetry, or similarity violations, and code-model MRs to possible determinism, rollout, or implementation issues. At this stage, this is a localization protocol, not a validated localization model. It becomes validated only if seeded faults or mutants with known layers are used to evaluate the inference rule. Section 5.3 reports a first bounded test: against an injected-fault catalogue the continuity

MR localized to boundary and normalization-scale faults while the symmetry MR localized to physical-channel and mesh-adjacency faults — suggestive evidence for the protocol's direction, but not, on one SUT and one catalogue, a validated localization model.

4. Empirical Design

4.1. Subject systems

The current evidence uses one trained MeshGraphNets-family implementation and checkpoint. The broader protocol names three intended implementation families, but they remain blocked until matched artifacts exist:

(1) an echowve MeshGraphNets PyTorch/PyG implementation;

(2) NVIDIA PhysicsNeMo `vortex_shedding_mgn`;

(3) DeepMind TF1 MeshGraphNets, or a third same-family configuration if runtime feasibility requires substitution.

For any future SUT admitted into the study, the experiment ledger must record repository URL, commit, checkpoint, dataset, mesh format, input fields, output fields, rollout horizon, random seeds, environment, and known runtime limitations before the manuscript can use it as evidence. Because these SUTs share a task and architecture family, even a completed cross-SUT package would be framed as a same-family stress test, not as evidence for all neural fluid surrogates.

4.2. Planned MR classes

The initial MR set will be grouped into representation MRs, geometric and symmetry MRs, continuity and constraint MRs, nondimensional similarity MRs, numerical stability MRs, and temporal or implementation MRs. Time reversal is excluded as a retained MR for viscous cylinder flow. Divergence is treated as an incompressible continuity or mass-conservation constraint, not as a Noether-derived conservation law.

4.3. Baselines and comparators

The study uses four comparator families.

Expert MR design. Domain experts manually propose MRs. This is the primary comparator because the proposed workflow is meant to make expert domain reasoning more explicit, reproducible, and executable.

Generic MR-generation scope contrast. Generic MR identification or generation methods are applied as far as their assumptions allow. This secondary comparator is not used to claim that generic methods are weak; it is used to show where domain-validity information is needed for SciML.

LLM-assisted candidate generation. An LLM generates candidate MRs from prompts containing equations, SUT descriptions, and candidate relation categories. The LLM is treated only as a candidate-generation and material-organization tool. It does not judge MR validity. Prompt logs, model/version, temperature, candidate lists, and rubric decisions will be recorded.

Rollout-accuracy diagnostic. Standard rollout error is reported to compare pointwise predictive evidence with relation-level evidence. The goal is not to prove that MRs are better than accuracy, but to identify whether they answer a different validation question.

Where feasible, residual-based or UQ metrics will be reported as diagnostic comparators. They will not be treated as MRs unless paired with a source/follow-up transformation and explicit verdict rule.

4.4. RQ-to-evidence map

Table 4 maps each research question to the planned evidence. This table is a pre-results design statement rather than a report of findings.

4.5. Efficacy parameters and statistical plan

The primary efficacy parameters are candidate retention rate, executable MR rate, MR violation rate, violation magnitude, fault-detection rate, boundary characterization, and interpretation yield. Secondary parameters include MR construction cost, inter-rater agreement for rubric decisions, flakiness rate across repeated executions, localization agreement with seeded-fault layers, and complementarity with rollout accuracy.

The current paper reports only one bounded within-SUT frame-level OOD-stress rate for mirror-y. Broader violation-rate estimates, confidence intervals, and paired comparisons remain blocked until the same source cases

Table 4: Research questions, efficacy parameters, baselines, and artifacts.

RQ	Primary parameters	Comparators	Artifacts
RQ1	candidate retention rate; rejection reasons; inter-rater agreement	expert MR design; LLM candidates	rubric sheets; candidate ledger; adjudication log
RQ2	executable MR rate; transformation success; asset completeness	expert MR cards; generic scope contrast	MR cards; transformation code; runner configs
RQ3	verdict distribution; out-of-domain rate; interpretation yield	residual/UQ diagnostics where available	verdict ledger; metric outputs; exclusion logs
RQ4	violation rate; violation magnitude; complementarity with rollout error; fault-detection rate if seeded faults exist	expert MR design as primary comparator; LLM/generic candidates as secondary contrasts; rollout accuracy as diagnostic evidence	run logs; plots; confidence intervals; fault ledger

and transformations are run across eligible SUTs or baselines. If those artifacts are later produced, binary verdict comparisons should use McNemar or Fisher-style tests only when assumptions are appropriate, and continuous violation magnitudes should use paired bootstrap intervals. When many MRs or transformation bins are compared, multiple-comparison correction or false-discovery-rate control must be reported. If the number of executable cases remains small, the study must emphasize effect sizes, confidence intervals, and qualitative failure interpretation rather than strong significance claims.

4.6. Ethics, integrity, and reproducibility

The current study does not involve human subjects, personal data, or private sensitive information. The main ethical and integrity issues are research transparency, AI use, and reproducibility.

All SUT versions, datasets, checkpoints, scripts, and run logs should be recorded when licensing permits. Failed, skipped, and inconclusive cases must remain in the experiment ledger. LLM use is restricted to candidate generation and evidence organization; it is not used as a final judge of MR validity. No candidate MR should be described as valid unless it passes the rubric and has a corresponding MR card. No OOD violation should be described as a program fault without the verdict evidence needed to support that interpretation.

Third-party code, datasets, and model checkpoints will be used according to their licenses. If any artifact cannot be redistributed, the paper will provide access instructions and record the limitation.

5. Results

Full cross-SUT, comparative, and fault-detection results remain blocked. This section reports only strictly scoped, within-SUT pilot evidence whose artifacts are committed and validated.

5.1. Claim-to-evidence map

The `PC#` identifiers are paper-level claims, kept distinct from the authoritative runtime claim ledger (`claim-ledger.yml`, claims `C1`–`C9`) so the two cannot be confused. The runtime claim ledger is the single source of truth for runtime-evidence claims; the mapping is PC1→C4, PC2→C1/C4, PC3→C3, PC4→C2, PC5→C7, PC6→C6, PC7→ no runtime-evidence claim (a method/ethics boundary), PC8→C8, PC9→C9, PC10→C10, PC11→C11, PC12→C12, and PC13→C13.

5.2. MR-card-to-verdict map

5.3. Within-SUT pilot evidence

All pilots run on one real trained MeshGraphNets cylinder-flow surrogate, using the read-only Minimum-MR-SubSet checkpoint recorded in the experiment ledger. They exercise three different rubric outcomes and are scoped accordingly; raw outputs, manifests, and metric ledgers are committed under `research_assets/runs/`.

Representation MR. Node-permutation equivariance holds to machine precision (relative L2 = 0.0 at tolerance 1e-6). This is a structural property of message-passing and is reported only as a pipeline sanity check, not as model capability.

Geometric MR. The rubric classifies exact mirror-y equivariance as out-of-relation-domain for this mesh: the reflection is non-bijective, the worst reflected-node mismatch is about one mesh edge length, and the cylinder is off-centre by 7.2 mm. The relation is therefore downgraded to an approximate nearest-neighbour OOD-stress probe scored by the predeclared MR-card metric against a same-space mapping-error floor. Within the same SUT and checkpoint, the approximate mirror-y OOD-stress probe failed on 10 of 10 recorded eval frames (median relative L2 0.737, median V/floor 3.96, 3.02–5.55x mapping-error floor range); every recorded frame is kept in the denominator and none was skipped or inconclusive. This is a bounded within-SUT frame-level OOD-stress result under an approximate reflection: one SUT, one checkpoint, one MR, one eval trajectory. It is not an exact mirror-symmetry result and not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim. The 10 frames are consecutive states of one trajectory and are not independent samples; an exact binomial 95% interval for 10 successes in 10 trials is wide ([0.69, 1.00]). The V/floor ratio is normalized by a mapping-error floor that is itself a function of the same geometric mismatch that triggered the downgrade, so on this asymmetric mesh V/floor cannot fully separate a model-level violation from an amplified geometric artifact; the exact-symmetry run on a symmetric mesh below resolves that ambiguity.

Continuity MR. A P1 discrete-divergence operator yields a non-negligible divergence even for the ground-truth field on this coarse mesh, so an absolute divergence-free tolerance is not calibratable and the absolute mass-conservation relation stays deferred. As a reference-relative diagnostic, the surrogate's predicted next-state divergence stays within about 0.4–0.8% of the reference on the recorded eval frames; the interior-only ratio confirms this is not only a boundary-imposition artifact. This asserts no absolute conservation. Two caveats bound the diagnostic: the reference divergence is not yet decomposed into discrete-operator error, solver projection artifact, or genuinely non-solenoidal training data, so if the reference field is itself materially non-solenoidal the reference-relative ratio is a non-regression guard rather than a conservation measurement; and the diagnostic uses a 50% regression threshold (ratio > 1.5) on two eval frames only, so "passes" means "does not regress conservation by more than 50% on those frames," not "conserves mass."

Rollout-accuracy diagnostic. On the same eval trajectory, the surrogate's one-step next-state prediction error ($v_{\text{pred}} = v_t +$ denormalized predicted delta, the trainer's own convention) has median relative L2 0.0216 (mean 0.044; min 0.0116, max 0.0788) over the nine recorded transitions. The per-transition error is bimodal (alternating $\sim$0.012 and $\sim$0.075), consistent with the vortex-shedding cycle, so we quote both statistics. The surrogate is accurate in-distribution to a few percent per step, yet the mirror-y OOD-stress violation (median 0.737) is larger by roughly an order of magnitude — about 34x the median accuracy and 17x the mean. Both quantities are dimensionless relative L2 but of different objects (equivariance of the model output versus next-state velocity error against the ground truth), so the ratio is an order-of-magnitude gap rather than a precise factor, and we do not compare the symmetric-mesh 1.10 to this number directly because their reference distributions differ. This is the first within-SUT evidence that the relation-level diagnostic and ordinary rollout accuracy answer different validation questions on this surrogate; it is a same-SUT accuracy diagnostic, not a baseline-superiority, multi-trajectory, or cross-SUT claim, and it is one-step, not a free-running rollout-stability result.

Exact mirror-y on a symmetric mesh. To test whether the mirror-y finding survives once the exact relation is admissible, we built a synthetic structured channel mesh that is provably symmetric about the centerline: the reflection is a verified bijection (node-type match 1.0, reflection offset $< 10^{-12}$, edge set invariant), so the admissibility predicate retains the exact relation rather than downgrading it. On one constructed input state the surrogate violated exact mirror-y equivariance with relative L2 1.10 (verdict fail). Because equivariance is oracle-free and structural — a mirror-equivariant model would satisfy it to machine precision regardless of accuracy — the nonzero result is not a geometric artifact, and it is an out-of-sample check that the admissibility predicate did not fit to the original three pilots. We ran one control to separate the learned weights from the input normalizer, which is fit to the asymmetric training data and so is not exactly equivariant in the transverse velocity: zeroing the normalizer's transverse-velocity mean (making it exactly equivariant in that component) changes the violation only from 1.1032 to 1.1014, so the normalizer accounts for about 0.2% of it and the violation is dominated by the learned message-passing weights, which carry no equivariance constraint. Two boundaries remain. The mesh is synthetic, has no obstacle, and is out-of-distribution for the cylinder-trained surrogate, so the *magnitude* 1.10 may be amplified by normalization mismatch relative to in-distribution behaviour; the result should be read as confirming that the surrogate carries no exact mirror-y equivariance constraint, while not bounding the in-distribution equivariance violation, rather than as a calibrated in-distribution magnitude. And this is one input on one mesh; it is not an accuracy, reliability, cross-SUT, or geometry-independent claim. The 1.10 magnitude is also larger than the asymmetric-mesh OOD-stress 0.737, which would be paradoxical if the symmetric mesh were the cleaner setting; the more likely reading is that the synthetic no-obstacle channel is itself more aggressively out-of-distribution for the cylinder-trained surrogate (Poiseuille-like profile rather than vortex shedding, no obstacle wake), so the larger magnitude reflects deeper OOD rather than a cleaner equivariance measurement. The two runs therefore answer different questions — admissibility of the relation, and severity of the violation under OOD — and the symmetric-mesh number should be read as a binary equivariance failure on an admissible relation, not as a directly comparable magnitude.

Seeded-fault detection. We re-implemented, in pure numpy/torch from the read-only Minimum-MR-SubSet witness taxonomy, a catalogue of 10 injected pipeline faults across five fault classes (boundary-condition, mesh-adjacency, normalization-scale, temporal-rollout, physical-channel), and used the paper's own MRs as detectors on the model's predicted update. The conservation (continuity) MR detected the two boundary-condition faults and the gross normalization fault (divergence ratio 3.8–10.6 vs the 1.5 threshold); the mirror-y (symmetry) MR detected a physical-channel and a mesh-adjacency fault (violation rising 69–142% above its 0.735 baseline); node-permutation equivariance detected none, because these faults preserve node-relabeling invariance and the MR stays exact by design. Five of ten mutants were detected by at least one MR (at least one mutant per detected fault class), and the detections localize by MR class — continuity to boundary/scale faults, symmetry to physical-channel/adjacency faults — which is a first bounded test of the interpretation protocol of Section 3.7, suggestive rather than a validation. Three honesty notes bound this. First, the detected faults are gross corruptions (zeroed inflow, non-zero wall, un-denormalized update, swapped velocity channels, permuted edges) that any divergence- or symmetry-sensitive detector would catch; the catalogue is an independent taxonomy, not adversarial to these MRs but not designed for them. Second, two undetected cases are near or by-construction: the edge-drop fault is a near-miss (32% mirror-y change vs the 50% threshold), and the boundary-condition faults are invisible to mirror-y by construction because boundary imposition happens downstream of the update the symmetry MR scores. Third, the remaining undetected faults (doubling a small update, sign-flipping the step, zeroing the transverse update) shift the absolute output without crossing the scored-quantity thresholds, delimiting where these MRs are structurally insensitive. It is one SUT, one checkpoint, one injected-fault catalogue; not a real-world or general fault-detection rate, a reliability claim, or a baseline-superiority claim.

5.4. Multi-checkpoint replication (K=6 checkpoints, one architecture family / one dataset)

To address a single-checkpoint objection without overclaiming a cross-SUT result, the five within-SUT measurements above are replicated across a roster of K=6 MeshGraphNets cylinder-flow checkpoints trained on the same DeepMind dataset using the same read-only Minimum-MR-SubSet trainer:

four base-config seed replicas S0–S3 (hidden=64, num_layers=4, training seeds $\{0, 1, 2, 3\}$, 1500 stage-A optimization steps each) and two configuration variants S4 (hidden=128, seed 0) and S5 (num_layers=6, seed 0). Each checkpoint has a distinct sha256 (recorded in its per-SUT manifest and in the E1 aggregate file under `multicheckpoint/`); the S4/S5 variants reuse the trainer's read-only `train()` entry point through a thin wrapper that writes only into this repository. The same six MR/diagnostic runners (node-permutation, mirror-y OOD-stress, exact mirror-y on the symmetric mesh, conservation diagnostic, rollout accuracy, seeded-fault detection) are executed against every checkpoint by the E1 orchestrator; the seed-replica family carries a bootstrap 95% confidence interval (B=2000 over S0–S3) and S4, S5 are reported descriptively as single-instance configuration variants. The honesty boundary is explicit: K=6 is a cross-checkpoint / cross-configuration replication across one architecture family and one dataset, not a cross-SUT (cross-architecture-family) generalization, and the seed-replica family standardises only the training seed and so should not be treated as an independent SUT sample. The replication results:

Node-permutation equivariance holds to machine precision (relative L2 = 0.0) on every one of the six checkpoints, confirming the representation-MR sanity check is a structural property of the message-passing pipeline rather than an artifact of one trained instance.

Mirror-y OOD-stress (approximate reflection on the asymmetric DeepMind mesh) reports a per-checkpoint median relative L2 with seed-family mean 0.774 and 95% bootstrap CI $[0.743, 0.804]$; S4 = 0.764 and S5 = 0.832 sit within or just outside that band. The violation is therefore reproducible across seed and configuration variation, not specific to one checkpoint.

Exact mirror-y on the synthetic symmetric mesh fails on every checkpoint: seed-family mean 1.00, 95% CI $[0.75, 1.15]$; S4 = 1.10, S5 = 1.00. The exact-symmetry failure is therefore a property of the architecture family on this dataset, not of one trained instance.

Reference-relative conservation ratio (max over the recorded eval frames) sits tightly above 1: seed-family mean 1.009, 95% CI $[1.007, 1.011]$; S4 = 1.008, S5 = 1.010. All six checkpoints stay well below the predeclared 1.5 regression threshold, so the reference-relative non-regression verdict reproduces across the family without any checkpoint passing absolute conservation.

One-step rollout accuracy (median relative L2) is also tight: seed-family mean 0.0221, 95% CI $[0.0217, 0.0224]$; S4 = 0.0226, S5 = 0.0221. The mirror-y OOD-stress violation is therefore about 35× the in-distribution one-step accuracy on *every* checkpoint of the roster, not just on the original pilot.

What this multi-checkpoint replication adds is not a new measurement but a robustness check: the within-SUT verdicts of Section 5.3 are not artifacts of a single training run. The mirror-y violation magnitude survives seed replication and mild configuration variation; the conservation diagnostic stays in its narrow band; the rollout accuracy is consistent. What this replication does *not* license is a cross-architecture-family generalization (only MeshGraphNets and a fixed dataset are exercised), an exact reliability claim (the seed-replicas share more variance than independent SUTs would), or a real-world fault-detection claim.

5.5. Operator-floor resolution sweep (calibration of numerical decidability)

The admissibility predicate of Section 3.3 requires the verdict tolerance to dominate the operator's intrinsic error floor (numerical decidability). For the P1 constant-per-cell discrete divergence operator that scores the continuity MR, that floor is a property of the operator and the mesh, not of any SUT, and it admits a direct empirical calibration. On the same $y = 0$-symmetric structured triangular channel mesh family used by the exact-mirror-y experiment of Section 5.3, at four resolutions $h_0, h_0/2, h_0/4, h_0/8$, we evaluated the analytically divergence-free reference velocity field $u = (\partial_y \psi, -\partial_x \psi)$ with stream function $\psi(x, y) = \sin(\pi x/L_x)\sin(\pi y/L_y)$ and measured the area-weighted RMS of the per-cell discrete divergence. A log–log linear fit of RMS vs. characteristic edge length gives slope **0.988** (all cells, $R^2 = 1.000$) and slope **0.995** (interior-only, $R^2 = 1.000$), matching the theoretical $O(h)$ truncation rate of P1 piecewise-linear shape functions on smooth fields. This empirically calibrates the operator-floor magnitude that the verdict tolerance must dominate, confirming the admissibility-predicate gating condition for the continuity MR has a measurable basis on this mesh family rather than only a theoretical justification. Raw outputs and the fit are committed under `operator-floor-sweep/`. The scope is one mesh family and one smooth analytic field; the calibration confirms the expected rate of the operator on this mesh family but does not generalize across mesh geometries or analytic fields.

5.6. Fault-detection robustness across the multi-checkpoint roster

To check whether the seeded-fault detection result of Section 5.3 reproduces beyond a single checkpoint and to delimit where its detectors are structurally insensitive, the 10-mutant catalogue is replayed against the K=6 multi-checkpoint roster across five input-permutation seeds (30 trials per mutant per detector), and two predeclared severity dimensions are swept: NS_double_scale $s \in \{1.1, 1.25, 1.5, 2, 4\}$ and PC_zero_vy partial fraction $p \in \{0.25, 0.5, 0.75, 1.0\}$ (the canonical mutant is $p = 1.0$). Detection is decided by the same predeclared thresholds as Section 5.3 (node-permutation tolerance 10^{-5}, conservation ratio threshold 1.5, mirror-y relative-change threshold 0.5). Wilson 95% CIs are computed across the (SUT, seed) cells. Raw outputs are committed under `fault-robustness-e3/`.

Per-mutant cross-family detection (R1). Four mutants are detected on every cell (30/30, CI $[0.89, 1.00]$): conservation flags both boundary-condition faults (BC_zero_inflow, BC_nonzero_wall) and the un-denormalized output fault (NS_skip_denorm); mirror-y flags the swapped-channel fault (PC_swap_xy). Five mutants are not detected on any cell (0/30, CI $[0.00, 0.11]$): MA_drop_edges, NS_double_scale, TR_sign_flip, TR_double_step, PC_zero_vy. The by-class localization of the original C10 pilot holds: continuity $\rightarrow$ boundary/scale faults, symmetry $\rightarrow$ physical-channel/adjacency faults, node-permutation $\rightarrow$ none of these mutants (they preserve node-relabeling invariance, and the representation MR stays exact by design). One mutant is configuration-sensitive: mirror-y detects MA_permute_edges on every (S0–S3, seed) cell (20/20 over the seed-replica family) but on neither (S4, seed) nor (S5, seed) cell (0/10 over the wider and deeper configuration variants). The within-family detection rate of that mutant is therefore 20/30 (CI $[0.49, 0.81]$) — a quantitative bound on within-family generalization that is invisible to a single-checkpoint experiment.

NS_double_scale severity sweep (R2). At every multiplicative scale $s \in \{1.1, 1.25, 1.5, 2.0, 4.0\}$ the detection rate stays at 0/6 (Wilson CI $[0.00, 0.39]$). This widens the bounded-insensitivity note from Section 5.3: multiplying the per-step output by a constant factor does not push the reference-relative conservation ratio past 1.5 (the per-step delta is small, so the next-state field stays close enough to the reference), does not move mirror-y by 50% (both the source and the mirrored output scale together, so the relative L2 between them is invariant under output scaling), and does not break node-permutation equivariance (a linear operation is permutation-equivariant). The MR catalogue at the current thresholds is therefore structurally insensitive to multiplicative output scaling on this surrogate across the swept grid, not only at the canonical 2× fault.

PC_zero_vy partial-fraction sweep (R3) — non-monotone detection. At partial zeroing fractions $p \in \{0.25, 0.5, 0.75\}$ the detection rate is 6/6 across the checkpoint roster (Wilson CI $[0.61, 1.00]$); at the canonical full fraction $p = 1.0$ the detection rate drops to 0/6 (CI $[0.00, 0.39]$). The intuition is structural: uniform $v_y = 0$ is itself mirror-y-symmetric ($0 = -0$ under reflection in the transverse component), so it preserves the very symmetry the MR scores; only spatially partial zeroing creates an asymmetric residual that the symmetry MR can resolve. The detection–severity curve is therefore non-monotone (Figure 1), with the canonical (full) fault sitting in a detection blind region that intermediate severities expose. This is a within-family finding about one MR–fault pair, not a general non-monotonicity claim across all symmetry MRs, but it is a concrete instance of a structural insight that a severity sweep makes explicit and a single-severity experiment hides: a fault MR may be insensitive to the most severe instance precisely because that instance shares an invariance the MR depends on.

Aggregate reading. The 5-of-10 union detection rate from C10 reproduces tightly across the seed-replica family (5/10 on each of S0–S3) and drops by one to 4/10 on the capacity (S4) and depth (S5) variants because MA_permute_edges crosses below the mirror-y relative-change threshold there. The by-class localization mapping (continuity → boundary/scale; symmetry → physical-channel/adjacency; node-permutation → none of these faults) replicates across the full roster. The catalogue's insensitivity regions — multiplicative output scaling at any swept severity, and a detection blind region at the most severe PC_zero_vy fraction caused by the fault's residual mirror-y symmetry — are now bounded with Wilson 95% CIs. This is a robustness and structural-insight result within one architecture family on one dataset against one fixed 10-mutant catalogue; it is not a real-world fault-detection rate, a cross-architecture-family generalization, or a baseline-superiority claim.

5.7. Still blocked

The following remain blocked and must not be written as results: cross-SUT or geometry-independent pass/fail rates; comparative superiority over any baseline; general or real-world fault-detection rates; localization accuracy as a validated model; runtime or performance claims; and any claim that one

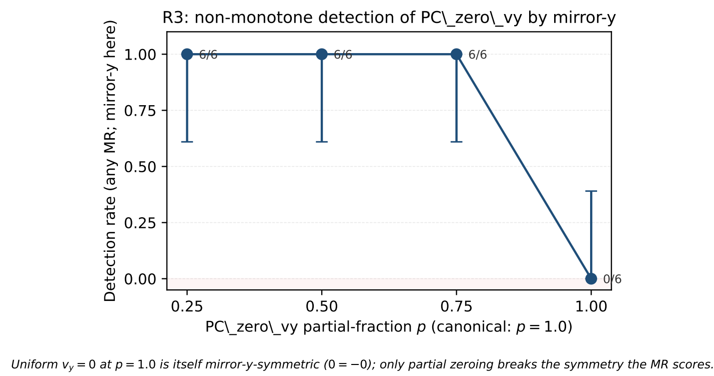

Figure 1: R3 non-monotone detection of PC_zero_vy by mirror-y. Partial-fraction zeroing $p \in \{0.25, 0.5, 0.75\}$ is detected on every checkpoint of the K=6 roster (6/6); the canonical $p = 1.0$ is detected on none (0/6). Uniform $v_y = 0$ is itself mirror-y-symmetric $(0 = -0)$, so the most severe fault sits in a detection blind region that intermediate severities expose. Error bars are Wilson 95% CIs across the 6-checkpoint roster.

SUT is more reliable than another. The three METBENCH-planned SUTs, and the expert-MR, generic-MR-generation, and LLM-candidate baselines, stay blocked pending their artifacts. Two exceptions are now executed on the existing SUT and reported as scoped diagnostics, not as defeated competitors or general rates: the rollout-accuracy comparator (Section 5.3) and a bounded seeded-fault detection result over one 10-mutant catalogue (Section 5.3), whose by-class localization is suggestive evidence for, not a validation of, the interpretation protocol. The multi-checkpoint replication of Section 5.4 is a cross-checkpoint / cross-configuration result over one architecture family and one dataset; it does not lift the cross-SUT block.

6. Discussion

6.1. Interpretation of the scoped evidence

The value of the current study is not that every MR finds a new fault beyond rollout accuracy. Rather, the value is that retained MRs provide relation-level evidence under explicit transformations. When a violation or deferral is observed, the MR card and verdict rule help distinguish model inconsistency, relation-domain boundary, numerical tolerance problem, and inconclusive evidence.

The PR4 mirror-y evidence adds one bounded rate claim only: the approximate OOD-stress probe failed on 10 of 10 recorded eval frames for one

SUT, one checkpoint, one MR, one eval trajectory. It remains an out-of-relation-domain exact mirror-y case and a bounded within-SUT frame-level OOD-stress result, not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim.

A natural objection is that the admissibility predicate merely re-describes the three outcomes it was introduced with — node permutation passes, mirror-y is downgraded, conservation is deferred — and is therefore fitted rather than tested. The two added within-SUT runs answer this. The symmetric-mesh run is an out-of-sample use of the predicate: it states that on a symmetric mesh the exact mirror-y relation becomes admissible, and the experiment then tests that admitted relation independently, finding a genuine equivariance violation (relative L2 1.10) rather than confirming a pre-arranged result. The rollout-accuracy run supplies the accuracy comparator the original three pilots lacked, showing the mirror-y violation is about 34 times the surrogate's in-distribution one-step error, so the relation evidence is not a restatement of accuracy. Neither run removes the central limitation — this is still one SUT and one checkpoint — but together they convert the mirror-y finding from a self-downgraded probe into an admissible-relation violation with an accuracy baseline.

The comparative baselines (expert MR design, generic MR generation, LLM candidates) remain blocked, so we cannot yet quantify what the rubric adds over unguided MR identification. We can, however, make the counterfactual concrete on the present evidence: the rubric averted two specific misreadings that an unguided application would plausibly have made. Without the numerical-decidability gate, the absolute discrete-divergence relation would have been executed and, because the surrogate's predicted divergence is close to the reference, recorded as a conservation *pass* — a false assurance, since the reference field itself carries non-negligible divergence and no calibrated tolerance exists; the rubric instead deferred it. Without the boundary-compatibility and bijectivity checks, the exact mirror-y failure on the asymmetric eval mesh would have been read as a model symmetry fault, when the symmetric-mesh run shows the exact relation is admissible elsewhere and the asymmetric-mesh violation is partly a geometric artifact; the rubric instead downgraded it to an OOD-stress probe and flagged the geometry. These are two concrete cases where the rubric prevented an overclaim the candidate relation alone would have produced — a counterfactual argument for the workflow's value, pending the blocked baseline measurements.

6.2. Boundary of claims

The paper must avoid four overclaims. It must not claim that MRs are always better than rollout accuracy. It must not claim that NOETHER proves the physical validity of cylinder-flow MRs. It must not claim that LLMs can automatically identify valid MRs. It must not generalize from three MeshGraphNets-family configurations to all SciML surrogates.

The safer claim is that a domain-validity-aware workflow can make MR identification and execution more auditable for a concrete class of SciML SUTs.

6.3. Implications for SciML testing

The scoped evidence shows how SciML testing can move from implicit expert checks to explicit MR assets. Such assets can complement residuals, uncertainty estimates, and accuracy metrics by making transformation assumptions and verdict rules inspectable. This is especially important for OOD validation, where the boundary of the relation is often as important as the violation itself.

In that sense the workflow is aimed at producing, over many controlled transformations, a relation-indexed applicability map: a record of where a surrogate stops respecting the relations it should respect, expressed in relation space rather than in residual space. The present evidence is one bounded within-SUT point on such a map, not the map itself; assembling a calibrated map would require the cross-SUT, multi-trajectory, and domain-violation-score work that remains future work.

7. Threats to Validity

Construct validity. MR validity depends on the rubric, physical assumptions, boundary-condition compatibility, and tolerance rules. Incorrect discrete operators or thresholds may create false violations or false passes.

Internal validity. SUT setup, checkpoint differences, random seeds, mesh preprocessing, and runtime nondeterminism may affect verdicts. The experiment ledger must record these details.

External validity. The empirical evidence covers one MeshGraphNets architecture family on one DeepMind cylinder-flow dataset, replicated across a K=6 checkpoint roster (four base-config training-seed replicas plus one wider and one deeper configuration variant; Section 5.4). Within that family the within-SUT verdicts are stable (CIs in Sections 5.4 and 5.6) and the operator floor that gates the continuity-MR admissibility predicate has the expected $O(h)$ rate on the symmetric mesh family (Section 5.5). The replication should not be read as a cross-architecture-family generalization: cross-SUT artifacts spanning distinct architecture families (e.g. PINNs, neural operators, other mesh-based simulators) remain blocked, and the seed-replica family standardises only the training seed, so it shares more variance than independent SUTs would. Generalization to all neural operators, PINNs, or fluid surrogates without further evidence is not supported.

Baseline fairness. Generic MR-generation and LLM baselines may not be designed for SciML. They should be interpreted as scope contrasts and candidate-generation comparators, not as defeated competitors.

Conclusion validity. Small sample sizes, multiple MRs, and many transformation bins can produce unstable conclusions. The mirror-y evidence has a bounded within-SUT frame-level OOD-stress rate, while conservation and node permutation remain pilots. Broader rates, external validity, seeded-fault effectiveness, reliability, accuracy, and baseline claims remain blocked.

Reproducibility. Some SUTs may depend on old runtimes or non-redistributable checkpoints. The paper should disclose such barriers and provide the most complete runnable package possible.

8. Conclusion

This paper presents domain-validity-gated MR identification as an auditable oracle-free testing workflow for MeshGraphNets-family cylinder-flow surrogates. The current evidence supports the scoped within-SUT pilots of Section 5.3 (node-permutation, mirror-y OOD-stress, conservation diagnostic, rollout-accuracy comparator, exact mirror-y on a symmetric admissible mesh, and seeded-fault detection over a 10-mutant catalogue), replicated across a K=6 multi-checkpoint roster of one MeshGraphNets architecture family on one dataset (Section 5.4) — the verdicts reproduce with tight Wilson and bootstrap 95% confidence intervals — together with an empirical calibration of the P1 discrete-divergence operator floor that gates the admissibility predicate (slope 0.988, $R^2 = 1.000$; Section 5.5), and a within-family fault-detection robustness study with severity sweeps that bound where the present MR set is structurally insensitive (Section 5.6). It does not support cross-architecture-family generalization, real-world or general fault-detection rates, baseline-superiority claims, reliability conclusions, accuracy conclusions, exact-mirror-y claims beyond the bounded results above, or absolute conservation claims. The central claim remains methodological: physically meaningful SciML MRs require explicit validity conditions, executable assets, raw evidence records, and relation-level verdicts; the K=6 replication and the operator-floor calibration are the present study's bounded evidence that the workflow's verdicts hold beyond a single checkpoint and that the admissibility-predicate gate it relies on has a measurable basis.

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Table 5: Compact claim-to-evidence map for the current manuscript.

Claim	**Status**	**Evidence**	**Boundary**
PC1-domain-validity-rubric	Method claim	Rubric asset and manuscript method section	Does not prove physical validity by itself.
PC2-mr-card-executable-assets	Workflow claim	MR cards and validators	Not every card has cross-SUT evidence.
PC3-baseline-comparison-blocked	Blocked	Claim and experiment ledgers	No baseline superiority, seeded-fault, localization, or runtime claim.
PC4-node-permutation-sanity	Observed pilot	Node-permutation metric ledger	One SUT and one pilot case only.
PC5-conservation-diagnostic-deferred	Diagnostic; absolute claim deferred	Conservation metric ledger and report	absolute conservation remains deferred.
PC6-mirror-y-ood-stress	Observed bounded pilot	Mirror-y rate metric ledger and claim ledger	failed on 10 of 10 recorded eval frames; not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim.
PC7-llm-candidate-support-only	Process boundary	Method and ethics sections	LLMs organize candidates; they do not judge MR validity.
PC8-rollout-accuracy-diagnostic	Observed	Rollout-accuracy metric ledger	Same-SUT one-step accuracy (median 0.0216); mirror-y is ∼34x larger; not baseline superiority.
PC9-exact-mirror-y-symmetric-mesh	Observed	Symmetric-mesh metric ledger	Exact relation admissible and fails (rel L2 1.10) on a synthetic OOD mesh; not accuracy/cross-SUT.
PC10-seeded-fault-detection	Observed	Seeded-fault metric ledger	MRs as detectors catch 5/10 injected mutants, localizing by MR class; not a general/real-world fault-detection rate.
PC11-multicheckpoint-replication	Observed	E1 aggregate and per-SUT manifests in `multicheckpoint/`	K=6 checkpoints of one MeshGraphNets family (S0–S3 seed replicas, S4 wider, S5 deeper); cross-checkpoint replication, not cross-architecture-family generalization.
PC12-operator-floor-resolution	Observed	Operator-floor sweep report	Log–log slope 0.988 (all)/0.995 (interior), $R^2 = 1.0$ over four resolutions; calibrates the admissibility predicate, not a mesh-family-independent claim.
PC13-fault-detection-robustness	Observed	E3 robustness report	30-trial Wilson CIs per (mutant, detector) across K=6; NS_double_scale undetected over $\{1.1, 1.25, 1.5, 2, 4\}$; PC_zero_vy non-monotone (partial detected, full undetected); MA_permute_edges configuration-sensitive. Within-family only.

Table 6: MR-card-to-verdict map for the three executed pilots.

MR card	**Rubric decision**	**Runtime verdict**	**Interpretation boundary**
Node permutation equivariance	Retained as representation MR	pass sanity; relative L2 $= 0.0$	Supports one executable representation path; does not establish model reliability.
Mirror-y equivariance (asymmetric eval mesh)	Exact relation out-of-relation-domain; downgraded to approximate OOD-stress	fail on 10 of 10 recorded eval frames; median relative L2 0.737; median V/floor 3.96	Bounded within-SUT OOD-stress violation for one trajectory; not by itself exact symmetry, cross-SUT, or geometry-independent evidence.
Mirror-y equivariance (synthetic symmetric mesh)	Exact relation admissible (bijection verified, offset $< 10^{-12}$, type-match 1.0)	fail; relative L2 1.10 on one input state	Exact-symmetry violation where the relation is admissible, removing the out-of-relation-domain objection; synthetic no-obstacle OOD mesh, one input.
Discrete divergence / conservation	Absolute mass-conservation MR deferred; reference-relative diagnostic retained	inconclusive: reference-relative non-regression guard on 2 frames (not scored as a conservation pass)	At 2 frames with a 50% slack threshold and an undecomposed reference divergence, this guards against regression but does not establish conservation.

Table 7: Multi-checkpoint replication aggregates over the K=6 SUT roster. CI is a B=2000 bootstrap over the seed-replica family S0–S3; S4 and S5 are reported as single-instance configuration variants.

MR / diagnostic (per-SUT scalar)	**S0–S3 mean**	**95% CI**	**S4**	**S5**
Node-permutation rel L2	0.0	$[0.0, 0.0]$	0.0	0.0
Mirror-y OOD-stress median rel L2	0.774	$[0.743, 0.804]$	0.764	0.832
Exact mirror-y on sym mesh rel L2	1.00	$[0.75, 1.15]$	1.10	1.00
Conservation max ratio	1.009	$[1.007, 1.011]$	1.008	1.010
One-step rollout median rel L2	0.0221	$[0.0217, 0.0224]$	0.0226	0.0221
Seeded-fault union (k/10)	5/10 (each)	—	4/10	4/10

Table 8: Per-mutant Wilson 95% confidence intervals from the E3 sweep across the K=6 multi-checkpoint roster and 5 input-permutation seeds (30 trials per mutant). "Detecting MR" is the MR whose threshold the mutant trips at its canonical severity.

Mutant	**Fault class**	**Detecting MR**	**k/n**	**Rate**	**95% CI**
BC_zero_inflow	boundary cond.	conservation	30/30	1.00	$[0.89, 1.00]$
BC_nonzero_wall	boundary cond.	conservation	30/30	1.00	$[0.89, 1.00]$
NS_skip_denorm	normalization	conservation	30/30	1.00	$[0.89, 1.00]$
PC_swap_xy	physical channel	mirror-y	30/30	1.00	$[0.89, 1.00]$
MA_permute_edges	mesh adjacency	mirror-y (S0–S3 only)	20/30	0.67	$[0.49, 0.81]$
MA_drop_edges	mesh adjacency	(none)	0/30	0.00	$[0.00, 0.11]$
NS_double_scale	normalization	(none)	0/30	0.00	$[0.00, 0.11]$
TR_sign_flip	temporal rollout	(none)	0/30	0.00	$[0.00, 0.11]$
TR_double_step	temporal rollout	(none)	0/30	0.00	$[0.00, 0.11]$
PC_zero_vy	physical channel	(none at $p = 1.0$)	0/30	0.00	$[0.00, 0.11]$